



Promoting the Culture of Academic and Research Integrity in a Higher Education Environment in Pakistan: A Qualitative Study

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Abstract

Given the evolution in the global educational landscape coupled with the emergence of various Artificial Intelligence tools, academic and research integrity has gained significant importance. This study aims to qualitatively explore the views of stakeholders including students and faculty members about the challenges to academic and research integrity and the possible strategies to strengthen the integrity culture in a higher education environment in Pakistan. The study used a qualitative approach by conducting a focus group discussion using pre-defined interview topics. Thematic analysis of the qualitative data led to generation of four major themes with respective subthemes. Inadequate support from the institute regarding awareness and resources, fear of failure, lack of plagiarism detection tools and quest for promotion were identified as contributing factors towards academic misconduct. Participants supported the role of continuous professional development in promoting academic integrity. Most of the participants suggested that institutes should take the responsibility of organizing awareness sessions about plagiarism and other forms of dishonesty. The findings of this study suggests that establishment of a culture of academic integrity and ethical research is challenging and requires a sustained effort to raise awareness and transform the attitudes and behaviors of students and researchers.

Keywords Academic integrity · Artificial intelligence tools · Plagiarism · Academic misconduct · Awareness · Ethical research

The study was ethically approved by the Research Ethics and Support Committee of the University of Management and Technology, Lahore, Pakistan (RESC-132–2023).

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Introduction

Given the evolution in the global educational landscape coupled with the emergence of various Artificial Intelligence tools, academic and research integrity has gained significant importance. Ethical academic behavior is derived from the combination of certain values such as fairness, honesty, trust, and respect. Academic integrity derived from these values creates a community dedicated to learning and the exchange of ideas (Holden et al., 2021). The term academic integrity is broadly used as a substitute for attitude of students and faculty, particularly in relation to plagiarism and deceptiveness (Macfarlane et al., 2014). There is a global increase in the incidence of academic dishonesty due to a gap between policies and practices within the institutions (Newton, 2018; Eaton et al., 2020). This critically affects the society, as dishonest students and faculty adopt mendacious behaviors in the organizations (Thomas & De Bruin, 2012). Promoting a culture of social responsibility, honesty and trust among the students and faculty can help prevent ethical misconduct in professional settings. A serious threat to these fundamental values in the form of fabrication, falsification, cheating and plagiarism raises a big concern for institutes and academicians (Geoffrey & Currie, 2023).

Unethical behaviors regarding research conduct were initially confronted in early 1990s and there was scarcity of sufficient empirical information that could identify possible causes (Rosenbaum, 1993). However, with advancement in research, expanded surveys comprising questions about motivation and attitude of researchers helped in understanding factors responsible for unethical research practices. This led to emergence of ideas to change research behaviors (Steneck, 2006).

Misrepresentation of research work is one of the forms of ethical misconduct whereby researchers fail to present contributions of all the authors accurately (Biswas, 2021). Furthermore, research indicates that researchers change sequence of authors, list unaccepted papers as “in press” and even might produce bogus publications (Dale et al., 1999; Baker & Jackson, 2000).

Another form of research misconduct is the inaccuracy and careless mistake, although some of research policies exclude them from unethical behavior (Steneck, 2006) in environments where people face extraordinary pressure to publish and funding procurement, careless mistakes in notes and bibliographies is not surprising (Gosling et al., 2004).

Fabrication, falsification and plagiarism (FFP) are three alleged academic frauds that disparagingly affect academic integrity. Plagiarism, though may not necessarily challenge the reliability of research, it may contribute to distrust among peers, wastage of time and funding. However, fabrication and falsification can have momentous influences on research. Intentional fabrication and falsification on part of researcher destabilizes trustworthiness of research record and relationships based on research (Steneck, 2006).

Academic misconduct is an increasingly prevalent issue in Pakistan with a number of new plagiarism cases of faculty members and students reported to their respective institutions and Higher Education Commission (HEC) of Pakistan. A study conducted to investigate academic integrity among medical students and postgraduate trainees in Pakistan reported that students engaged in cheating and plagiarism to cope with various assessments and evaluations (Azam & Naeem, 2022). Another study conducted by Soroya et al. (2016) suggested that a number of students plagiarize due to lack of knowledge regarding citation norms and they use cheating as a shortcut to success to escape from the social and familial

pressure to achieve high grades. Given the reported high prevalence of academic misconduct in Pakistan coupled with the lack of awareness and inadequate institutional mechanism to uphold academic integrity, there is a dire need to gain deeper insight into the perceptions, experiences and the challenges faced by the key stakeholders. Addressing this gap is essential for designing the possible strategies that could be implemented to foster the culture of academic and research integrity in a higher education environment in Pakistan. This study therefore, explored a range of integrity issues faced by the academic institutions such as assessments related misconduct including cheating in exams, plagiarism and undue authorship demand in assignments as well as research-related infractions including fabrication, data falsification and unethical authorship in publications.

The aim of the study is to qualitatively explore the perspectives of stakeholders including the faculty members and students regarding academic and research integrity with a focus on the challenges they encounter and the effective strategies that could strengthen the culture of academic and research integrity within the academic institutions of Pakistan. Our study explicitly seeks to answer the following research questions:

1. What are the factors that contribute to academic misconduct?
2. What role does the Continuous Professional Development (CPD) play in promoting integrity culture?
3. What are the barriers and challenges that the students and faculty members face while upholding academic integrity?
4. What are the possible strategies that can help strengthen the culture of academic and research integrity in higher education institutions in Pakistan?

Methods

This study used a qualitative approach by conducting a focus group discussion using pre-defined interview topics (please see appendix 1). Focus group discussion provides diverse perspectives and insights of the faculty and students regarding the phenomenon (Patton, 2014).

The study was conducted at the School of Pharmacy of a large chartered private university in Lahore, Pakistan. The study was ethically approved by the Research Ethics and Support Committee of the University of Management and Technology, Lahore, Pakistan (RESC-132–2023). The RESC arranges workshops and seminars focused on academic ethics and integrity. These sessions are aimed to raise awareness among faculty and students about academic integrity and conduction of research as per the four principles of biomedical ethics including autonomy, beneficence, non-maleficence, and justice. Components of ethically informed consent and the importance of scientific peer review in research were also emphasized in these seminars to help the participants improve their understanding and compliance with the highest principles of academic integrity and research. Purposive sampling (Palinkas et al., 2015) was used for the study whereby the faculty members and students who attended workshops and seminars arranged by the RESC were invited to participate in the focus group discussion voluntarily. The discussion was recorded and written informed consent was taken from the participants. None of the participants (both faculty and students) held any formal responsibility regarding ethical misconduct. Reflections from students and

faculty were based on the awareness they had received from workshops organized by RESC as well as from the challenges they faced in the institutional environment.

The focus group discussion lasted approximately 60 min. The discussion was moderated by a senior faculty member who had been involved in arranging RESC workshops and was trained in qualitative interviewing. Care was taken to keep mixed composition of students and faculty participants. Moreover, the moderator ensured equal opportunities to speak and respectful dialogues throughout the discussion. All the participants were encouraged to openly share their perspectives. Audio recorded data of focus group discussion with the faculty members and the students was transcribed verbatim. The participants were assured about the data privacy that it would only be accessible to the authors of the study. An inductive method of thematic analysis using Braun and Clarke's (Braun & Clarke, 2006) six-step framework was conducted. Researchers transcribed the data and developed categories and themes from the recording. The thematic analysis was carried out by three researchers (SR, EC, ZA). Initially, the researchers independently reviewed the data. Further, a collaborative discussion was made after the independent review of the researchers. The data categories were condensed to themes and the responses listed under each theme were examined for agreement or disagreement, avoiding any kind of biasness. To ensure the trustworthiness of the data, a thick and rich description was used (Cohen et al., 2002). The trustworthiness of study was ensured by following Lincoln and Guba's framework (credibility, confirmability, dependability and transferability) (Alexander, 2019). The credibility of the study was maintained through prolonged engagements, follow-up discussions to further explore and verify the themes related to the objectives of the study. Confirmability was ensured by keeping the records of coding decisions while making sure that the interpretations were based on participant's exact words. Dependability was maintained by keeping a clear research protocol, field notes and avoiding inconsistencies throughout the study. Transferability was maintained by providing rich description of participant's views allowing the readers to have an understanding of the findings to other contexts. The use of think-rich description enables us to explain the actual meaning behind the perspective and address the issues of generalizability of the results.

Results

The focus group discussion included three senior faculty members (Participant #2, 3 and 4) and five students (Participant # 1, 5, 6, 7 and 8). Please see appendix 2 for the participants' demographics. Thematic analysis of focus group discussion led to generation of four major themes with respective subthemes. These along with the supporting quotes from participants are presented in Table 1.

Theme 1: Factors Contributing To Academic Misconduct

Inadequate guidance from the institute, fear of failure and increasing grading standards, lack of plagiarism detection tools and quest for promotion were identified as contributing factors towards academic misconduct.

Table 1 Themes, sub-themes and supporting quotes

Themes	Sub-themes	Supporting quotes
Factors contributing to academic misconduct	Inadequate institutional guidance	Participant 1 (Student) "We have never been guided about plagiarism. The only thing we understood while composing any assignment was to copy the content from resource and paste it as such."
	Limited Plagiarism Detection	Participant 2 (Faculty Member) "No one from my institute ever guided about research ethics or the importance of ethical approval forms."
	AI-Generated Content	Participant 3 (Faculty Member) "I confronted rejections from multiple journals while struggling to publish my data due to lack of guidance from supervisor."
	Fear of failure and increasing grading standards	Participant 3 (Faculty Member) "Institutes have not incorporated sufficient AI detection capabilities in Turnitin service to find out academic dishonesty done with the help of emerging AI writing tools."
	Quest for Promotion	Participant 4 (Faculty Member) "Our Turnitin accounts should be upgraded to latest versions which incorporate efficient means to detect use of AI writing tools by students."
		Participant 2 (Faculty Member) "With the emergent AI tools, the incidence of falsification, plagiarism and fabrication has intensified."
		Participant 5 (Student) "A few members in study group ask for unauthorized help from us with the fear of failure".
		Participant 1 (Student) "Low performers in our class, who have fear of failure ask for unauthorized help from other group members."
		Participant 2 (Faculty Member) "Occasionally students are overburdened that triggers the practice of academic dishonesty to avoid failure."
		Participant 3 (Faculty Member) "Students struggle with the pressure to maintain their cGPA. Requirement to procure superior grades instigates cheating and falsification."
		Participant 5 (Student) "It is nice to help peers but sometimes it is difficult to facilitate the lazy fellows who contribute nothing in group assignments but demand reward in terms of grades because they do not want to fail. I think this is academic dishonesty and we all should avoid it."
		Participant 6 (Student) "We need to maintain anticipated cGPA to sustain the merit scholarships but this seems difficult with reformed grading trends. We got our marks deducted in assignments due to plagiarism as this is not always easy to change statements of scientific findings."
		Participant 4 (Faculty Member) "Competition and aspiration for lucrative publications to secure academic promotion and appraisals damages academic integrity."
		Participant 2 (Faculty Member) "Academic integrity is damaged due to desire of beneficial article publication to achieve promotion."

Table 1 (continued)

Themes	Sub-themes	Supporting quotes
Role of CPD in promotion of academic integrity	Contents delivery and relevance in training program Impact of CPD on Faculty and Students	Participant 1 (Student) "The resource person conspicuously elaborated the concepts of research ethics and academic integrity. Contents of slides were remarkable."
		Participant 2 (Faculty Member) "I got an idea what plagiarism is as well as the effective strategies to avoid it."
		Participant 4 (Faculty Member) "For me, the session on academic integrity was very much interactive and engaging. It was not as if the speaker has just come to speak and we are here to eavesdrop on. Rather he interacted substantially with the participants that further elaborated the understanding about falsification and plagiarism."
		Participant 3 (Faculty Member) "The knowledge in the slides was well communicated by the speaker with real life examples. Even he displayed the format of research ethics approval form in detail. Many researchers don't know about it while contriving research strategy."
		Participant 6 (Student) "I was really impressed by the style and motivational directives of speaker. We could not expect that someone could disseminate such kind of knowledge so amicably."
		Participant 7 (Student) "I was so happy that the contents were relevant and self-explanatory. I consistently applied it to correct the recurring mistakes in my writings."
		Participant 1 (Student) "The seminars helped me in improving the quality of my assignments"
		Participant 2 (Faculty Member) "The CPD seminars were close to my individual learning needs and preferences. I am a freelancer and write for some international magazines. These seminars were focused on the learning component and I gained knowledge about effective writing protocols. Now, as I have learned about plagiarism, I feel more confident while writing."
		Participant 3 (Faculty Member) "I learned to minimize plagiarism while utilizing citations and quotations as evidence because this was an obstacle while inscribing any document. Facilitators expounded the ways to minimize plagiarism while presenting scientific evidences and quotes."
		Participant 4 (Faculty Member) "Students who attended CPD seminars display discernable improvement in their writings regarding clarity and quality of contents."

Table 1 (continued)

Themes	Sub-themes	Supporting quotes
Barriers and challenges	Lack of awareness and resources	Participant 1 (Student) "I don't know how to manage plagiarism. I face difficulty in rephrasing some scientific statements and knowledge that hinders to remain within allowed plagiarism limit. Professors might think its deliberate but how can I observe the confined plagiarism limit when I lack basic training."
	Cultural barriers	Participant 2 (Faculty Member) "Our society lacks awareness about misconduct and integrity. There is no sufficient guideline in the form of seminars and workshops that may be provided to make us aware of it".
	Peer pressure to publish	Participant 3 (Faculty Member) "We have limited resources to conduct research that deprives us from giving a broader spectrum to research ideology."
		Participant 4 (Faculty Member) "As a faculty member, I must have access to latest plagiarism detection services that may trace misconduct by use of artificial intelligence."
Way forward		Participant 2 (Faculty Member) "Multiple barriers such as gender discrimination, family duties, time limitations, lack of family support and distrust amongst colleagues develops poor research culture that contributes towards misconduct."
		Participant 1 (Student) "Being a female, I have multiple tasks to justify. It is not always possible for me to make all the ends meet. There is pressure from family and lack of encouragement and support."
		Participant 1 (Student) "High publication pressure for career advancement compromises research integrity."
		Participant 2 (Faculty Member) "People use shortcuts to publish that weakens the credibility of scientific work."
		Participant 3 (Faculty Member) "Request from peers for co-authorship is an emotional issue that gets difficult to handle at times."
		Participant 1 (Student) "Awareness sessions encompassing seminars and workshops should be conducted in institutes that deliver meticulous guidelines about academic integrity and research ethics."
	Awareness sessions	Participant 2 (Faculty Member) "Awareness sessions should incorporate such activities that offer hands on practice to remove plagiarism."
	Motivation	Participant 1 (Student) "Institutes should provide a motivational atmosphere that promotes quality research and discourage falsification."
	Resources	Participant 3 (Faculty Member) "Institutes should propose awards or incentives to employees who publish quality research work"
	enhancement	Participant 4 (Faculty Member) "Students should be motivated to critically review and reflect on their self-produced documents and be counselled to overcome the fear of failure."
		Participant 2 (Faculty Member) "Institutes should ensure availability of resources in terms of funding and access to paid journals to ensure that researchers do not engage themselves in any scientific misconduct".
		Participant 4 (Faculty Member) "Efficient plagiarism detection tools should be made available to faculty."

Inadequate Institutional Guidance

Majority of the participants identified scarcity in provision of ethical guidance from the institutes as a major contributing factor towards academic misconduct. The participants observed that early career researchers often lack the mentorship and meticulous support from their affiliated institution. This intentional or unintentional inadvertence compromises academic integrity. Furthermore, participants highlighted that there is a dire need of dissemination of knowledge regarding best ethical practices in research (please see Table 1 for supporting participants' quotes).

Limited Plagiarism Detection for AI-Generated Content

The faculty participants shared their concern over the increasing misuse of AI writing tools in academia. They were of the view that AI writing tools are being utilized by students that provide rephrased data without citations. They further highlighted that the currently available plagiarism detection tools such as Turnitin lack effective AI detection capabilities and stressed the need for upgradation of these tools to better identify academic misconduct such as falsification and plagiarism.

Fear of Failure and Increasing Grading Standards

Majority of the student participants identified the fear of failure and pressure to maintain their cGPA and secure scholarships as one of the contributing factors to academic misconduct. Participants reported that direct or indirect influence from peers who secure high grades compels towards academic dishonesty to achieve good grades at par with their peers. This also increases their anxiety levels and prompts them to engage in the act of misconduct. A few also reported frustration with group work, where peers do not contribute yet expect shared credit to avoid failing.

Quest for Promotion

Faculty participants highlighted the quest for promotion as another contributory factor towards academic misconduct and compromised ethical standards in research and publication practices. Participants observed that the pressure to publish quickly for academic promotion and annual appraisals challenges academic integrity. Furthermore, academic institutions consider publications in high impact factor journals as essential for promotion.

Theme 2: Role of CPD in Promotion of Academic Integrity

Participants highlighted the positive role of Continuous Professional Development (CPD) including the relevance of contents delivery in training programs as well as the role of facilitators in promoting academic integrity.

Contents Delivery and Relevance in Training Program

Participants had a positive opinion about their experience and perception regarding the contents and their delivery in training program. Similar to the content quality and delivery, narratives of participants regarding content relevance to CPD theme were quite promising.

Impact of CPD on Faculty and Students

Illustrations from faculty and students about CPD impact depicted gains that could be attained by purposeful CPD programs and highlighted the importance of need analysis for participants before designing any CPD program.

Theme 3: Barriers and Challenges

The participants enthusiastically communicated the barriers and challenges that result in moral disengagements such as plagiarism, fabrication and other similar forms of academic dishonesty.

Lack of Awareness and Resources

Majority of the students' participants reported that before attending CPD sessions, they had no idea about plagiarism. According to them lack of awareness about the rules results in poor adherence to academic integrity. Participants blamed the academic institutions for not providing enough training opportunities to avoid academic misconduct. Similar, observations were made by the faculty participants who identified the non-availability of sufficient research opportunities and resources as a source of unintentional academic misconduct (please see Table 1).

Cultural Barriers

Some of the participants articulated numerous cultural barriers including gender biasness, assumptions and lack of trust among peers promotes dishonesty and negatively impact the research culture. These challenges, particularly for women, create pressure and hinder productivity, which can contribute to academic misconduct.

Peer Pressure To Publish

The participants were of view that they have to endure a lot of pressure in academic setups to publish in high impact factor journals in order to secure annual appraisals and promotion. This inculcates an unintentional deviation from ethical conduct. It might be even in the form of unauthorized acknowledgement in others work.

Theme 4: Way Forward

Awareness Sessions

Based on the participant's retorts about various factors that challenge academic integrity and research ethics, they were further asked for the pointers that could be supportive in curtailing academic dishonesty. Most of the participants had a view point that it is the affiliated institute that should take the responsibility of organizing awareness sessions about plagiarism and other forms of dishonesty.

Motivation

The participants emphasized the significance of motivation to improve academically harmful dishonest mindsets and conducts.

Resources Enhancement

Both students and faculty members advocated the need of resources enhancement to instigate quality research culture in institutes (please see Table 1 below).

Discussion

To authors' knowledge, this is the first focused group qualitative study in Pakistan that is based on the outcomes of focus group discussion about factors that cause academic dishonesty and way forward to impede it in academics and research. Thematic analysis of focus group discussion led to findings that there are numerous factors that impact academic integrity.

According to participants, inadequate support from the institutes regarding awareness and resources, fear of failure, lack of plagiarism detection tools and quest for promotion serve as the major contributing factors towards academic misconduct. Previous qualitative studies on reasons of academic dishonesty have shown that fabrication and falsification are being done by scarcity of institutional guidelines, impersonation from family or friends, need to perform better to avoid failure and issues of morality (Henning et al., 2014).

In Pakistan, there are a number of institutional, systemic and cultural challenges due to which academic integrity is potentially compromised in higher education sector. Despite the increasing awareness about plagiarism in academic institutions, academic dishonesty in the form of cheating and plagiarism is quite prevalent. Although Higher Education Commission (HEC) of Pakistan has made plagiarism checks mandatory through tools including Turnitin, there seems to be more focus on the similarity index rather than the context of matching content. Dishonest behaviors in academic institutions could be avoided only by critical evaluation, education and training of both faculty and students (Soroya et al., 2016). Limited understanding of academic ethics, coupled with lack of clarity and weak enforcement of academic policies lead to dishonest behaviors including plagiarism, contract cheating and collusion. Although tools like AI plagiarism checkers are available, educational reforms and ethical trainings are mandatory to promote the culture of academic integrity. In

countries like Pakistan, people work together in groups in form of collectivism, however, in academic institutions, this habit causes confusion as they are unable to differentiate between legitimate collaboration and collusion (Ali et al., 2024).

Participants highlighted the positive role of continuous professional development (CPD) in the promotion of academic integrity. Participants suggested that those seminars had given them awareness about plagiarism and other forms of academic misconducts and were close to their learning needs and preferences. They accredited that the pragmatic style of facilitators improved their writing skills and confidence. From reported findings, it has been estimated that institutes, preferably those who offer postgraduate studies try to prevent academic dishonesty by arranging awareness seminars, student's training on scholarly writing and utilization of antiplagiarism and AI writing tool detector software. Researchers have ascertained the importance of training to strengthen the academic integrity. This might be further augmented by use of student university portals and social media platforms (Maina et al., 2023; Money, 2008). Continuous development programs (CPD) are integral in improving content, procedural and conceptual knowledge along with the skills and attitudes of researchers. This enables them to strengthen professional bases and educate other learners more effectively (Luneta, 2012).

Participants identified few barriers that might prevent the individuals to follow honest behaviors in learning environments. Stakeholders of focus group discussion articulated numerous cultural barriers including gender biasness, lack of awareness and resources, assumptions and lack of trust among peers that most likely compel the students towards dishonesty. These findings respond to our research question regarding the challenges to foster the integrity culture. It has been established that competition among peers and drive towards higher scientific productivity offer a number of consequences that might promote dishonest attitudes. A higher number of publications and citations is an important contributing factor in individual and institutional reward. The resultant pressure fosters a sense of dishonesty and falsification to achieve more with minimum effort and time (Van Dalen & Henkens, 2012). Prioritization of academic success over academic integrity leads to misconduct (Çelik & Razi, 2023).

In line with the research question, the participants not only highlighted the challenges but also shared their way forward to foster academic integrity. Most of the participants demanded institutional support in terms of enhanced awareness, resources and motivation to suppress dishonest behaviors of researchers. It has been concluded earlier that institutional support is the key factor to promote the value of integrity. Institutes should guide about the policies and procedures to follow academic integrity with the involvement of faculty members (Gallant & Drinan, 2006). Enhancing the problem-solving skills and critical thinking of the students and learners must be articulated to promote the culture of academic integrity (Steneck, 2006). The stakeholders recommended that institutes should offer short courses on academic integrity. Researchers have reported feedback from different institutes whereby it has been suggested that course on academic integrity should be a part of curriculum (Maina et al., 2023).

Study Limitations

The current study offers a few limitations. The perceptions and behaviors of students and faculty members as well as the ethical policies may vary from institute to institute. The

study was conducted within a single institution. A more generalized study can be carried out involving multiple institutions. The study was conducted on a small sample size. This could be increased further to get more generalized results. The study also lacked comparative data from various institutes. A more comprehensive study with multiple institutes would also help to attain comparative data. Although efforts were made to create a non-judgmental and open forum for taking views of both faculty members and students, the students may have felt inhibited in expressing views in the presence of faculty members. This could have possibly influenced the openness and depth of student's responses.

Conclusion

This study signifies the priorities of academic institutions in promoting the culture of academic and research integrity. Guided by the research questions, the findings underline both the challenges faced by the students and faculty regarding academic ethics and the way forward to strengthen the integrity culture in educational institutes. The academic institutes should regularly review, update and implement their academic integrity policies to address the evolving ethical challenges in academia such as extensive use of AI and contract cheating. For a shared understanding of academic integrity, awareness sessions should be organised regularly for both students and faculty. The assessments methods should be further diversified to discourage the unethical practices of cheating, plagiarism and fabrication. The promotion of academic and research integrity practices is challenging and requires a sustained effort to transform the attitudes and behaviors of students, faculty and researchers.

Appendix 1

Interview topics	Questions
1. Motivation	→ What motivated you to attend the seminar on academic integrity?
2. Content and delivery	→ How did you find the delivery of the seminar, including presentation slides, structure and facilitator? → Do you believe the seminar's content is relevant to your academic discipline or research area? → Were there specific examples that resonated with your own experience?
3. Impact on Behavior	→ What are the factors that contribute to academic misconduct? → Do you feel that attending the seminar has influenced your ethical conduct in your academic or research activities? How so? → Have you implemented any of the principles discussed in the seminar in your academic or research work? If yes, please share your experiences and any challenges you encountered. → Have you encountered any barriers or challenges in applying the principles of academic integrity and research ethics in your academic or research practice? If so, please describe them.
4. Further support and resources	→ What kind of support or resources would be helpful to you in maintaining a commitment to academic integrity and research ethics?
5. Way forward to minimize academic misconduct	→ What suggestions do you have for improving academic integrity and research ethics?

Appendix-II

Participant	Designation	Gender	Age
Participant 2	Associate Professor	Female	45 years
Participant 3	Assistant Professor	Female	40 years
Participant 4	Assistant Professor	Male	40 years
Participant 1	Student	Male	20 years
Participant 5	Student	Female	19 years
Participant 6	Student	Male	19 years
Participant 7	Student	Female	20 years
Participant 8	Student	Female	20 years

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Data Availability All data has been included in the manuscript.

Declarations

Ethics Approval The Study has been ethically approved by the Research and Ethics Support Committee of the University of Management and Technology Lahore, Pakistan.

Consent Informed written consent has been obtained from all the participants.

Competing interests None.

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